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6.2mm × 6.2mm light emitting area on 12mm × 12mm emitter footprint.

Coloured LED chips

City farming or vertical farming uses a cultivation process where crops are grown in small interior areas using LED arrays to provide an optimal light spectrum. This is done by mixing LEDs of various colours for boosting growth throughout the year, irrespective of how the Sun is shining.

The problem of beam halos and mismatched, unmixed colours is claimed to be resolved here by using one focal length for all colours. Cree announced XQ-E high-intensity LEDs that deliver maximum candelas through secondary optics, enabling lighting manufacturers to radically reduce system size and power of high-intensity colour applications.

Adding further, the company release mentions that it is built on Cree's SC5 technology platform and features a combination of optical symmetry, consistent design across all colours, and a tiny 1.6mm × 1.6mm footprint simplifies manufacturing and design while providing excellent colour mixing.

Investing in longevity

Lighting products started off as gas-filled tubes with inverters, which were then followed by compact fluorescent lamps. "Components that go in LED lighting products today dissipate comparatively less heat and do not use any chemical gases," explains G. Jeganathan, director, GTP Equipments India Pvt Ltd. However, heat still affects the reliability of LEDs by a significant amount.

"The brand of the component is not really connected to quality. Any single component failure will be a cost to the company, and that hurts its image. In our case, we have set up complicated test equipment and systems to make sure every LED that goes through us is tested with different current variations, switching frequencies and environmental conditions," says Joy Biswas, product manager, Forbix Semicon.

Heat is definitely a problem for these devices. "Bulbs that come with aluminium heat-sinks do a good job of dissipating heat. The more heat these dissipate, the more reliable are the bulbs. This is because excess heat can damage the internals. For example, copper wiring is affected by continuous heat over a period of time, which can end up tearing the transformer if too much heat resides inside the unit," says Biswas. Now that you know this, consider the fact that quite a few inexpensive LED bulbs come with plastic heat-sink components!

What next

"In the future, lumens per watt will improve at least five to ten times over conventional LEDs, while cost per watt will dramatically come down. Multi Junction Technology (MJT) LEDs are more reliable than conventional LEDs today," says I. Hari Prasad from OMTRONICS LED Lighting Solutions. Engineers using MJT LEDs will be able to eliminate drivers or converters in their designs. Prasad also explains that MJTs have a longer life and enjoy low-distortion power by having one of the lowest total harmonic distortion along with the highest power factors. ●

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